

MATH 1342 – Elementary Statistics – Week 7: Confidence Intervals and the Bootstrap

Oct 6 – 8 | 3.2 Understanding and Interpreting Confidence Intervals • 3.3 Constructing Bootstrap Confidence Intervals Using Standard Error | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#
3. Find and use empirical probabilities in bootstrap distributions to find confidence intervals and in randomization distributions to test hypotheses. 9. Estimate single means, difference of two means, single proportions and difference of two proportions using confidence intervals. Interpret the results.	1. Construct a 95% confidence interval for a mean or proportion as statistic $\pm 2 \cdot \text{SE}$, given the statistic and its standard error. (CO9) 2. Interpret a confidence interval in context, explain the confidence level as the long-run capture rate of the method, and identify common misinterpretations. (CO9) 3. Describe how a bootstrap sample and a bootstrap distribution are created, and explain why the bootstrap distribution is centered at the sample statistic. (CO3) 4. Use StatKey to generate a bootstrap distribution and estimate the standard error as its standard deviation. (CO3) 5. Construct and interpret a 95% bootstrap confidence interval using the standard-error method. (CO3, CO9)	e-Text: 3.2 Understanding and Interpreting Confidence Intervals	MO1, MO2	Participation: Interpreting Confidence Intervals – which of 20 simulated CIs capture μ ; fix three misinterpretations	MO1, MO2	Online Homework: 3.2 Understanding and Interpreting Confidence Intervals (due Sun Oct 11, 11:59 PM)	MO1, MO2
		Interactive Lecture: 3.2 Understanding and Interpreting Confidence Intervals	MO1, MO2	Participation: Bootstrap a Proportion – Try It: 32 of 50 prefer Brand A; build the bootstrap distribution, read off SE, form the CI	MO3, MO4, MO5		
		Slides: 3.2 Understanding and Interpreting Confidence Intervals	MO1, MO2			Online Homework: 3.3 Constructing Bootstrap Confidence Intervals (due Sun Oct 11, 11:59 PM)	MO3, MO4, MO5
		e-Text: 3.3 Constructing Bootstrap Confidence Intervals	MO3, MO4, MO5				
		Interactive Lecture: 3.3 Constructing Bootstrap Confidence Intervals	MO3, MO4, MO5				
		Slides: 3.3 Constructing Bootstrap Confidence Intervals (embedded StatKey simulator)	MO3, MO4, MO5			Test 1 Retake – Mon Oct 5 – Sun Oct 11 (optional; higher score counts)	None
		StatKey (lock5stat.com/statkey) – Bootstrap Confidence Intervals	MO4, MO5				